

Molecular profiling of glioblastoma: spatial comparison against control brain tissue

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```
library(omicsdata)
library(SummarizedExperiment)
library(tidyverse)
```

1 Biological context and study objectives

2 Experimental design

3 Data and preprocessing

3.1 Data loading and brief look at it

```
if (file.exists("SRP044668.rds")) {
  SRP = readRDS("SRP044668.rds")
} else {
  stop("dear user: the necessary data is not present in the current working directory or
       does not have the right name, please check.")
}
```

Let's give a look at the data's structure.

```
#str(SRP)
dim(SRP)
```

```
## [1] 63856    94
```

```
assays(SRP)
```

```
## List of length 1
## names(1): raw_counts
```

3.2 Subsetting

We are now going to subset the data coherently with our goals.

```
sample_info = colData(SRP)$sra.sample_attributes

# Using new columns in the ColData I classify the columns (samples) of the
# dataset based on the type.

# 3-way classification

colData(SRP)$class_3way =
  case_when(grepl("brain tissue", sample_info)~"control",
            grepl("contrast-enhancing", sample_info)~"glioma_CE",
            grepl("non-enhancing", sample_info)~"glioma_NE")
```

```
# 2-way classification
```

```
colData(SRP)$class_2way =  
  case_when(grepl("brain tissue", sample_info)~"control",  
            grepl("glioma", sample_info)~"glioma")
```

3.3 Unexpressed genes removal

As a team, we decide to exclude genes with expression levels at 0 or really low levels in all samples here, before the EDA, and not in the quality control paragraph. We do this because we believe this row filtering is not a quality control but a basic definition of the rows we consider “data”.

```
as.data.frame(table(rowSums(assay(SRP, "raw_counts")) == 0))
```

Var1	Freq
FALSE	54034
TRUE	9822

```
as.data.frame(table(rowSums(assay(SRP, "raw_counts")) < 10))
```

Var1	Freq
FALSE	53885
TRUE	9971

```
as.data.frame(table(is.na(assay(SRP, "raw_counts")))) #good!
```

Var1	Freq
FALSE	6002464

```
as.data.frame(table(duplicated(rownames(SRP)))) #good!
```

Var1	Freq
FALSE	63856

```
SRP = SRP[rowSums(assay(SRP, "raw_counts")) > 10, ]
```

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